
Reinforcement Learning from Human Feedback (RLHF) with CartPole

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1 Introduction

Reinforcement learning (RL), through trial and error, mimics how humans learn from experience. It is a powerful machine learning technique, used to train software to make optimal decisions. In RL, taking action results in feedback to "reward" or "punish" certain actions with the goal of learning a policy that will maximize reward over time. RL has been utilized to train algorithms to play complex games like chess and Go as well as optimize how robots and autonomous vehicles interact with their environment. In spite of its great potential, however, RL faces several significant challenges that are preventing it from being widely adopted in more real-world applications. One of these is that RL algorithms struggle with stability and convergence, often producing unpredictable or inconsistent results. Addressing this issue is important for advancing RL beyond the specific tasks it has been employed in and allowing it to be deployed to solve complex real-world problems.

In particular, certain tasks are easily evaluated by a human but are tough to explicitly define and so remain challenging for machine learning models to optimize. Does a text sound biased? Is the media being watched entertaining? A human can assess this readily and rank "phrase one" as sounding more biased than "phrase two". Humans can also readily score and rank the entertainment value of "game 1" versus "game 2" or provide a numerical assessment of how much each was enjoyed. This human feedback can be leveraged in machine learning models to more efficiently train models, rather than having to, for example, curate large datasets of biased and non-biased texts. This is the paradigm that underpins reinforcement learning from human feedback (RLHF), a technique that incorporates human preferences into model optimization.

Human-machine interaction in machine learning holds significant potential, revolving around the collaborative relationship between a program and its human user. Ideally, this interaction should be highly cooperative and user-friendly. However, an inherent challenge is that human users often exhibit suboptimal and unpredictable behaviors, with their preferences difficult to articulate or quantify precisely. This raises a key question: how can a program be designed to optimize an unknown reward function that reflects a human user's preferences? Our approach leverages reinforcement learning and online training to customize user-specific adaptations of the algorithm based on individual user behavior and preferences. In a controlled experimental setup, these algorithms generate simulations for the game CartPole, rated subjectively by users based on their perception of the simulation. CartPole is a version of the pole-balancing problem and was introduced by Richard Sutton in his 2011 IEEE paper as a problem that is crucial in understanding how brains learn difficult concepts; it accomplishes this task through mimicry of neuron-like adaptive elements. Our project aims at unraveling how subjective rating introduces unpredictable, suboptimal feedback to the CartPole problem. Success is defined by increasing user satisfaction ratings across training iterations, as the algorithm adapts to generate different playing strategies based on user ratings. This study explores the extent to which reinforcement learning can capture and influence user-specific, implicit reward functions in cooperative human-machine interactions.

2 Summary of data

Data collected in this study consisted of subjective user ratings from interactions with individualized versions of the algorithm, each tailored to capture specific user preferences. For each training round, users rated generated simulations within the CartPole arcade environment on a subjective scale, reflecting the degree of interest or aesthetic satisfaction they experienced. These ratings served as feedback for the respective algorithm instances, guiding their iterative adjustment to better align with each user’s implicit preferences. The dataset includes these ratings across multiple training rounds for each user-algorithm pair, capturing variations in user satisfaction as the algorithm progressively optimizes pattern generation. Additionally, metadata such as training iteration numbers, pattern characteristics, and individual user identifiers were recorded to facilitate analysis of both algorithmic and user-specific trends in preference learning.

3 Methods

This study will perform comparative reinforcement learning with human feedback implementing a Bayesian approach for the reward function with optimization. We will train this model traditionally to play CartPole and optimize reaching the win state as fast as possible. We will compare this with the objective of maximizing human satisfaction which will be subjectively rated on a normalized scale. This study employed an online version of reinforcement learning algorithm with human feedback (RLHF) to develop personalized human-machine interaction models. The RLHF algorithm was adapted to iteratively learn from user feedback within a cooperative framework, focusing on optimizing an unknown reward function representing implicit user preferences. To modify this framework, we will use explicit human feedback to guide and shape the learning process. Each user interacted with an individualized instance of the algorithm, which generated simulated gameplay in the CartPole arcade environment. Users rated these gameplay patterns on a subjective scale according to their personal interest in the resulting output. The algorithm updated its parameters based on these ratings, with each iteration aiming to enhance its ability to produce simulations that aligned with each user’s preferences. Data on user ratings, pattern characteristics, and algorithm iterations were collected and used to evaluate the effectiveness of the model in capturing and adapting to the user’s subjective preferences over time. The success of each algorithm instance was measured by an increase in user satisfaction scores across training rounds, indicating improved alignment with the user’s implicit reward function.

4 Preliminary Results

Preliminary results demonstrate a successful implementation of a reinforcement learning algorithm to play the game of CartPole on a user-friendly interface in which the states can be visualized so that the human user can subsequently rate the performance of the algorithm in terms of how subjectively satisfying it is. In the current implementation, the algorithm is optimizing reaching the win state in a human-observable manner as its policy goal. The modified algorithm will incorporate the human feedback to alter the policy for optimization of subjective human satisfaction.

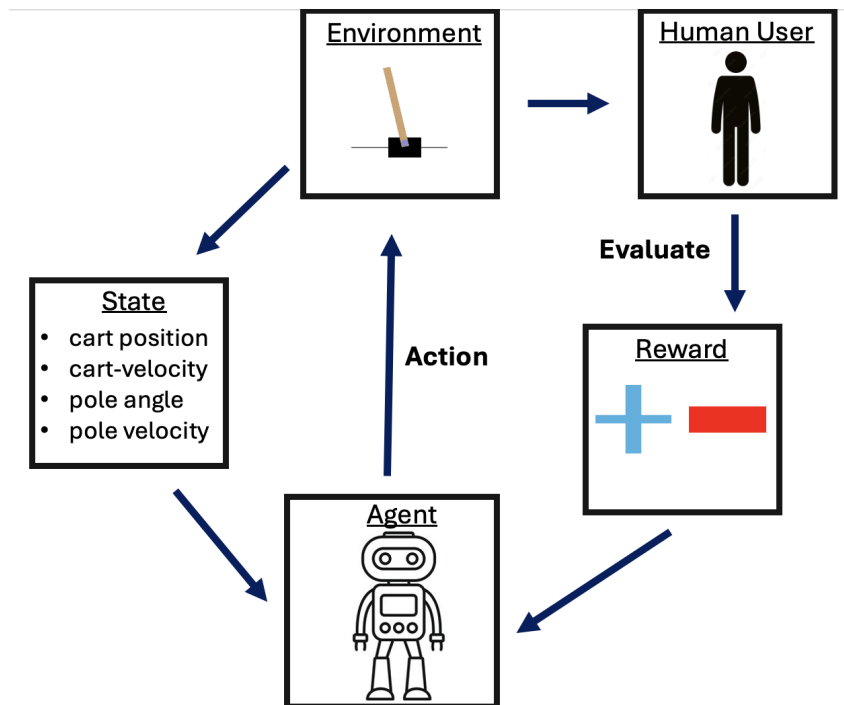


Figure 1: Approach to CartPole RLHF Implementation

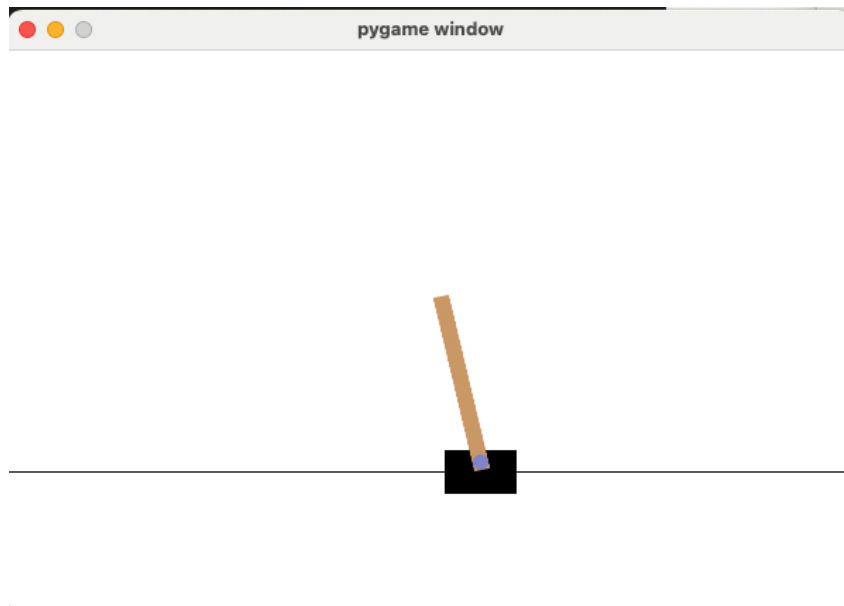


Figure 2: User Interface for Visualizing CartPole Being Played